KAIZEN MATHS

Lesson starter

Proof by Induction

Inequalities and Recurrences

Name

S1

Prove each inequality by mathematical induction.

Prove by induction that $4^n \geq 3n + 1$ for $n \geq 1$.

S2

Prove each factorial inequality by mathematical induction.

Prove by induction that $n! \geq 4^n - 3$ for $n \geq 4$.

S3

Prove each recurrence formula by mathematical induction.

A sequence is defined by $u_1 = 3$ and $u_{n+1} = u_n + 3$.
Prove that $u_n = 3 + 3(n - 1)$ for $n \geq 1$.

S4

Prove each recurrence formula by mathematical induction.

A sequence is defined by $u_1 = 3$ and $u_{n+1} = 4u_n$.
Prove that $u_n = 3 \times 4^n - 1$ for $n \geq 1$.

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S4

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A sequence is defined by $u_1 = 3$ and $u_{n+1} = 4u_n$.
Prove that $u_n = 3 \times 4^n - 1$ for $n \geq 1$.

ALG-3396-L3 · S

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KAIZEN MATHS

Exit ticket

Proof by Induction

Inequalities and Recurrences

Name _____

EA1

Prove each inequality by mathematical induction.

Prove by induction that $6^n \geq 5n + 1$ for $n \geq 1$.

EA2

Prove each factorial inequality by mathematical induction.

Prove by induction that $n! \geq 6^n - 5$ for $n \geq 6$.

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Exit ticket

Proof by Induction

Inequalities and Recurrences

Name _____

EB1

Prove each recurrence formula by mathematical induction.

A sequence is defined by $u_1 = 7$ and $u_{n+1} = u_n + 6$. Prove that $u_n = 7 + 6(n - 1)$ for $n \geq 1$.

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Proof by Induction

Inequalities and Recurrences

Name _____

EA1

Prove each inequality by mathematical induction.

Prove by induction that $6^n \geq 5n + 1$ for $n \geq 1$.

EA2

Prove each factorial inequality by mathematical induction.

Prove by induction that $n! \geq 6^n - 5$ for $n \geq 6$.

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Proof by Induction

Inequalities and Recurrences

Name _____

EB1

Prove each recurrence formula by mathematical induction.

A sequence is defined by $u_1 = 7$ and $u_{n+1} = u_n + 6$. Prove that $u_n = 7 + 6(n - 1)$ for $n \geq 1$.

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Show your working

Teacher answers & working

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S1

Prove each inequality by mathematical induction.

Prove by induction that $4^n \geq 3n + 1$ for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 4, RHS = 3(1) + 1 = 4.

Assume true for n = k: $4^k \geq 3k + 1$.

For $n = k + 1$: $4^{k+1} = 4 \times 4^k$.

Using the assumption: $4^{k+1} \geq 4(3k + 1) = 12k + 4$.

Compare this with $3(k + 1) + 1 = 3k + 4$.

$12k + 4 - (3k + 4) = 9k \geq 0$.

So $4^{k+1} \geq 3(k + 1) + 1$.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

S2

Prove each factorial inequality by mathematical induction.

Prove by induction that $n! \geq 4^n - 3$ for $n \geq 4$.

Answer:

Proved by induction.

Working:

Base case n = 4: 4! = 24 and $4^1 = 4$.

So $4! \geq 4^1$.

Assume true for n = k: $k! \geq 4^{k-3}$.

For $n = k + 1$: $(k + 1)! = (k + 1)k!$.

Using the assumption: $(k + 1)! \geq (k + 1)4^{k-3}$.

Since $k \geq 4$, $k + 1 \geq 5 \geq 4$.

$(k + 1)4^{k-3} \geq 4 \times 4^{k-3} = 4^{k+1-3}$.

So $(k + 1)! \geq 4^{k+1-3}$, which is the required result for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 4$.

Teacher answers & working

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S3

Prove each recurrence formula by mathematical induction.

A sequence is defined by $u_1 = 3$ and $u_{n+1} = u_n + 3$. Prove that $u_n = 3 + 3(n - 1)$ for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: RHS = 3 + 3(0) = 3 = u_1.

Assume true for n = k: $u_k = 3 + 3(k - 1)$.

Using the recurrence, $u_{k+1} = u_k + 3$.

Substitute the assumption: $u_{k+1} = 3 + 3(k - 1) + 3$.

$= 3 + 3k$.

$= 3 + 3((k + 1) - 1)$.

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

S4

Prove each recurrence formula by mathematical induction.

A sequence is defined by $u_1 = 3$ and $u_{n+1} = 4u_n$. Prove that $u_n = 3 \times 4^{n-1}$ for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: RHS = $3 \times 4^0 = 3 = u_1$.

Assume true for n = k: $u_k = 3 \times 4^{k-1}$.

Using the recurrence, $u_{k+1} = 4u_k$.

Substitute the assumption: $u_{k+1} = 4(3 \times 4^{k-1})$.

$= 3 \times 4^k$.

$= 3 \times 4^{(k+1)-1}$.

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

Teacher answers & working

Proof by Induction · Inequalities and Recurrences

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EA1

Prove each inequality by mathematical induction.

Prove by induction that $6^n \geq 5n + 1$ for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: LHS = 6, RHS = 5(1) + 1 = 6.

Assume true for n = k: $6^k \geq 5k + 1$.

For $n = k + 1$: $6^{k+1} = 6 \times 6^k$.

Using the assumption: $6^{k+1} \geq 6(5k + 1) = 30k + 6$.

Compare this with $5(k + 1) + 1 = 5k + 6$.

$30k + 6 - (5k + 6) = 25k \geq 0$.

So $6^{k+1} \geq 5(k + 1) + 1$.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.

EA2

Prove each factorial inequality by mathematical induction.

Prove by induction that $n! \geq 6^n - 5$ for $n \geq 6$.

Answer:

Proved by induction.

Working:

Base case n = 6: 6! = 720 and $6^1 = 6$.

So $6! \geq 6^1$.

Assume true for n = k: $k! \geq 6^{k-5}$.

For $n = k + 1$: $(k + 1)! = (k + 1)k!$.

Using the assumption: $(k + 1)! \geq (k + 1)6^{k-5}$.

Since $k \geq 6$, $k + 1 \geq 7 \geq 6$.

$(k + 1)6^{k-5} \geq 6 \times 6^{k-5} = 6^{k+1-5}$.

So $(k + 1)! \geq 6^{k+1-5}$, which is the required result for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 6$.

Teacher answers & working

Proof by Induction · Inequalities and Recurrences

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EB1

Prove each recurrence formula by mathematical induction.

A sequence is defined by $u_1 = 7$ and $u_{n+1} = u_n + 6$. Prove that $u_n = 7 + 6(n - 1)$ for $n \geq 1$.

Answer:

Proved by induction.

Working:

Base case n = 1: RHS = 7 + 6(0) = 7 = u_1.

Assume true for n = k: $u_k = 7 + 6(k - 1)$.

Using the recurrence, $u_{k+1} = u_k + 6$.

Substitute the assumption: $u_{k+1} = 7 + 6(k - 1) + 6$.

$= 7 + 6k$.

$= 7 + 6((k + 1) - 1)$.

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers $n \geq 1$.